

Prompt Framing Lab Teacher Notes

TOK Cognitive Science Activity Bank • Knowledge and Technology • 30-40 min

Category	Knowledge and Technology	Time	30-40 min
Difficulty	Medium	ID	prompt-framing-lab

Big idea: A prompt is not just a request; it frames what kind of knowledge output is likely.

Overview

Students compare how different prompts can make the same AI-style task produce different answers and assumptions.

Student Prompt

Inspect Prompt Framing Lab Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Prompt Framing Lab Stimulus A	Student-facing stimulus 1: Neutral prompt: 'Explain whether homework improves learning.'. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Prompt Framing Lab Stimulus B	Student-facing stimulus 2: Leading prompt: 'Explain why homework clearly improves learning and list supporting evidence.'. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Prompt Framing Lab Reveal / Twist	Student-facing stimulus 3: Leading prompt: 'Explain why homework clearly improves learning and list supporting evidence.'. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — A prompt is not just a request; it frames what kind of knowledge output is likely.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Prompt Framing Lab Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Prompt Framing Lab Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Who is responsible for knowledge produced through a prompt?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how a prompt is not just a request; it frames what kind of knowledge output is likely.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Use teacher-written mock outputs if live AI access is not appropriate.
- Keep the task constant while changing only the prompt wording.

Ready-to-Use Examples

- Neutral prompt: 'Explain whether homework improves learning.'
- Leading prompt: 'Explain why homework clearly improves learning and list supporting evidence.'

Suggested Timing

Stage	Teacher move	Watch for
1	Give groups the same knowledge task with different prompt wordings.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Students predict how their prompt might shape the answer.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Groups compare generated or mock outputs for tone, evidence, uncertainty, and omissions.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Students revise one prompt to improve reliability and transparency.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief how human framing and automated generation interact.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- The output begins before the machine answers; it begins with the frame we give it.
- What did your prompt make easy for the system to say?

Common Misconceptions

- Students may treat a tool's output as neutral because it feels automatic.
- Students may focus only on user error and miss design incentives or hidden rules.

Assessment Look-Fors

- Identifies the rule, design choice, or incentive shaping knowledge.
- Distinguishes access to information from justified knowledge.

Differentiation and Adaptations

- Support: compare two prompts only.
- Challenge: add a source-checking phase where students verify one output claim.

Extension Tasks

- Students write a prompt policy for using AI in TOK research.
- Compare prompt framing with survey wording and loaded headlines.

Related Activities

- AI Hallucination Courtroom
- Autocomplete Poetry
- Search Engine Ranking Game